

Detecting Cross-Area Concept Links in HCI Using Persistent Homology

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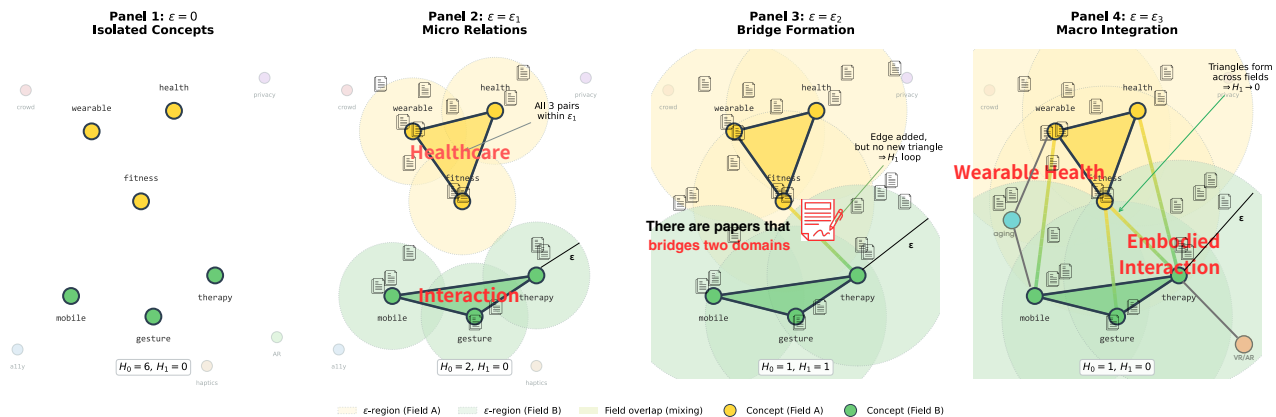


Figure 1: A Vietoris–Rips filtration illustrating how persistent homology detects cross-area bridges. As the proximity threshold ϵ increases, isolated concepts form local clusters (panels 1–2), then a cross-community edge creates a structural loop (panel 3), and finally triadic closure collapses the loop, signaling field integration (panel 4). The H_1 loop in panel 3 represents a semantic gap—adjacent but uncoupled research areas—that persists until direct co-occurrence becomes common.

Abstract

Identifying emerging research connections is essential for tracking how research fields evolve, yet many bibliometric methods rely on co-occurrence counts and therefore emphasize links only after they become common. We apply topological data analysis to 26 years of HCI literature (approximately 30,000 papers from CHI, UIST, CSCW, and UbiComp). For each year, we build a similarity-based concept complex from embeddings and compute persistent 1-cycles: rings of related concepts that remain unfilled, for example A–B–C–D–A with no diagonals, indicating semantic proximity without triangle closure. We cluster concepts into communities and treat cross-community edges appearing on a representative cycle as candidate “bridges,” while returning intermediate concepts for

human inspection. We observe cycles that persist for more than a decade. Moreover, cycle death often precedes a sharp rise in direct co-occurrence between endpoints in subsequent years, typically by one to three years. We interpret these events as signals of structural readiness for integration that complement frequency-based metrics.

CCS Concepts

• **Information systems** → Digital libraries and archives; • **Human-centered computing** → HCI theory, concepts and models; • **Mathematics of computing** → Topology.

Keywords

topological data analysis, research trend analysis, persistent homology, science of science in HCI

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1 Introduction

Research advances through connections between concepts from different areas. In HCI, significant developments often occur when concepts from separate domains intersect [16]. Identifying emerging cross-domain connections is important for understanding how research fields evolve, yet detecting these signals early remains challenging. Prior bibliometric approaches—frequency-based methods (co-occurrence networks [8], topic models [1]) and similarity-based methods—focus on strong, existing signals. Tools like CiteSpace [3] and VOSviewer [8] visualize established themes through accumulated citation data, providing retrospective rather than prospective analysis. Because these methods rely on dense patterns, they can miss weaker, early precursors to collaboration. This challenge is exacerbated in HCI, where core concepts are continuously contested [22], making simple proximity signals hard to interpret.

We address these limitations using Topological Data Analysis (TDA), specifically persistent homology, which captures global structural patterns beyond local similarity. Unlike prior work using topology as a numeric feature [4, 6], we use it to provide inspectable explanations via representative cycles. Conceptually motivated by boundary objects [24]—concepts bridging intersecting communities—we look for persistent H_1 features: loops of related concepts in which shortcut connections are missing. Intuitively, a persistent loop indicates two areas moving closer through intermediates but not yet directly linked, revealing structural gaps before co-occurrence accumulates. Prior work has applied PH to co-occurrence networks [26], framed knowledge gaps [27], and used topological features for link prediction [28]. We differ by using *semantic embeddings* (enabling earlier signals), focusing on cross-community bridges in HCI, and providing inspectable explanations via representative cycles. We applied this method to 26 years of HCI publications (approximately 30,000 papers from CHI, UIST, CSCW, and UbiComp) to address the following research questions.

Research Questions. (RQ1) Do persistent H_1 cycles predict cross-area links before co-occurrence? **(RQ2)** Does cycle death correspond to subsequent integration in the literature?

Contributions. We contribute: (1) a topological method using persistent H_1 cycles to identify cross-area bridges before co-occurrence accumulates; (2) empirical evidence that persistent loops (10–20 years) often close 1–3 years before co-occurrence spikes; and (3) representative cycles as inspectable explanations, namely, chains of intermediate concepts that make suggestions traceable rather than opaque.

2 Methodology

2.1 Data Pipeline and Topological Analysis

Data was collected from CHI, UIST, CSCW, and UbiComp proceedings (1999–2025, 1999–2024 for CSCW), comprising approximately 30,000 papers. Concepts were extracted from titles and abstracts using noun-phrase parsing with spaCy [11]; multi-word phrases (e.g., “virtual reality”) are joined with underscores before embedding to preserve phrase semantics. We filtered stopwords and rare

concepts (appearing in fewer than five papers) to focus on established sub-communities. We define the **Historical Corpus** $\mathcal{P}_{\leq t}$ as the set of all papers published in or before year t . All processing for year t uses only $\mathcal{P}_{\leq t}$ to prevent future information leakage. For semantic embedding, we trained year-specific Word2Vec models [19] (dimension $d = 100$, window size 5, minimum count 5) on $\mathcal{P}_{\leq t}$. While higher dimensions might capture more semantic nuance, they also increase the sparsity of the space, potentially creating artificial topological holes. We found $d = 100$ to be a robust trade-off for capturing macro-level structural relationships. We use a persistence threshold $\tau = 0.05$ in all experiments. After filtering, the raw vocabulary can be several thousand terms; for evaluation we keep the top 500 most frequent concepts per year to keep the H_1 computation tractable. For H_1 we compute the full pairwise cosine distance matrix and build the Vietoris–Rips complex directly (maximum simplex dimension 2). Separately, we build a weighted k -NN graph ($k = 15$) for baselines, candidate sets, and neighbor statistics. Let $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ denote the set of concepts, where each concept v_i has a Word2Vec embedding $\mathbf{v}_i \in \mathbb{R}^d$ ($d = 100$). The distance matrix $D \in \mathbb{R}^{N \times N}$ is defined as $D_{ij} = 1 - \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \cdot \|\mathbf{v}_j\|}$, i.e., cosine distance (1 minus cosine similarity). In practice, many concept vectors are semantically related, so distances often fall in a narrower range than the theoretical maximum. For each distance threshold ϵ , we construct a Vietoris–Rips complex $\mathcal{R}_\epsilon$: a simplicial complex where a k -simplex exists if and only if all pairwise distances among its $(k + 1)$ vertices are $\leq \epsilon$. As ϵ increases, we obtain a filtration $\mathcal{R}_{\epsilon_1} \subseteq \mathcal{R}_{\epsilon_2} \subseteq \dots$ with $\epsilon_1 < \epsilon_2 < \dots$.

Persistent homology quantifies this evolution. Each H_1 generator γ is characterized by birth threshold $\epsilon_b(\gamma)$ and death threshold $\epsilon_d(\gamma)$. Persistence is $\pi(\gamma) = \epsilon_d(\gamma) - \epsilon_b(\gamma)$. Algebraically, H_1 captures 1-dimensional cycles that lie in the kernel of the boundary operator on edges but are not themselves boundaries of filled triangles (2-simplices). We retain H_1 features with $\pi(\gamma) \geq \tau = 0.05$ as robust signals, and we observed stability across $\tau \in [0.035, 0.075]$. In this setting, a persistent H_1 feature can be interpreted as a semantic hole: a ring of related concepts surrounding a region of the embedding space where no concept currently resides, suggesting a potential missing connection in the discourse.

2.2 Detecting Structural Gaps

For each candidate pair $(u, v) \in \mathcal{C}_t$, H_1 Bridges assigns a score:

$$s(u, v) = \max_{\gamma: (u, v) \in E(\hat{\gamma})} \pi(\gamma), \quad (1)$$

where $\hat{\gamma}$ is the reconstructed representative cycle for generator γ . Pairs that do not appear on any reconstructed cycle receive $s(u, v) = 0$, and ties are broken by cosine similarity. We operationalize yearly structural gaps as total persistent H_1 mass:

$$\text{Gaps}(t) = \sum_{\gamma \in H_1(\mathcal{R}^{(t)}), \pi(\gamma) \geq \tau} \pi(\gamma). \quad (2)$$

The sum extends over all detected 1-cycles in the global complex $\mathcal{R}^{(t)}$, regardless of community boundaries, capturing system-wide structural tension in the concept space. We use the term “loop stability” to refer to $\pi(\gamma)$ in a given year; longer persistence indicates a more robust structural gap. We define cross-area links

using concept groups. We identify groups with the Louvain algorithm [9] on a separation graph G_{sep} built at ϵ_{sep} , defined as the largest threshold before the largest connected component exceeds 30% of nodes. We rely on these data-driven communities rather than fixed taxonomies (like ACM CCS) because HCI subfields are fluid and rapidly evolving. "Cross-area" in this paper therefore refers to bridging structurally distinct clusters in the semantic space, which may or may not map one-to-one with administrative subcommittees. We use Louvain because it does not require pre-setting the number of groups and works well on weighted graphs at this scale. Community assignments are $C : V \rightarrow \mathbb{Z}_+$. We treat a persistent H_1 loop γ (a cycle that survives across thresholds) as a cross-area link candidate when: (1) its persistence satisfies $\pi(\gamma) \geq \tau = 0.05$, and (2) the loop includes at least one edge (u, v) with $C(u) \neq C(v)$. We reconstruct the loop by taking the birth edge (u, v) and the shortest path between u and v in the graph just before birth. All cross-group edges on that loop are reported as cross-area links.

3 Evaluation Study Design

We evaluate early cross-area link discovery on cross-community, non-adjacent pairs with no prior co-occurrence ($c_{\text{max}} = 0$). For base year t , with vocabulary V_t and Louvain communities C on $G_{\text{sep}}(t)$, the candidate set is defined as

$$C_t = \{(u, v) : u < v, u, v \in V_t, C(u) \neq C(v), c_{\leq t}(u, v) \leq c_{\text{max}}\}, \quad (3)$$

where $c_{\leq t}(u, v)$ denotes the co-occurrence count of concepts u and v up to year t . Each method scores all pairs $(u, v) \in C_t$ and returns the top- L pairs, where $L = 50$. Integration uses a first-hit criterion within a horizon H ; we set $H = 6$ for ranking evaluation and $\theta = 5$ as the co-occurrence threshold. We use time-decay NDCG@ L with gain $g_i = \#[\text{integrated}] / \log_2(1 + \Delta t_i)$, where Δt_i is years until first hit; earlier integrations yield higher gain. Lead time is measured using Kaplan–Meier curves with right-censoring at $t + H$. We limit the vocabulary to the top 500 terms per year for stability. All baselines operate on the same candidate set C_t to ensure to a fair comparison. Baselines include Semantic $\times$ Under-cooccurrence, Triangle Deficit, Adamic–Adar combined with Resource Allocation (graph-based link prediction), Edge Betweenness, and Louvain cross-community cosine ranking. We omit embedding-based graph link predictors that typically require dense neighborhoods, and compare methods on the same sparse cross-community candidate set C_t . H_1 links are projected onto C_t , and pairs outside the candidate set are discarded.

We evaluate across 10 rolling windows (2010–2019) with $H = 6$, requiring data through 2025. Statistical procedures include: (1) a weight-shuffling null model (100 permutations) for topological structure validation; (2) bootstrap confidence intervals (1,000 iterations) for mean integration rates; (3) permutation tests (10,000 permutations, BH FDR $\alpha = 0.05$) for method comparisons. Sensitivity to ϵ_{sep} (largest component at 20/30/40% of nodes) is reported in Section 4. The corpus and preprocessing pipeline are described in Section 2. To ensure temporal validity, the method at year t uses only publications published in or before year t . To validate interdisciplinaryity, we cross-referenced concepts with ACM CCS. H_1 boundary pairs connect distinct high-level CCS categories at about $2\times$ the baseline rate ($p < 0.001$), stable across $k \in \{10, 15, 20\}$.

Weight-shuffling nulls show observed total persistence well below the null expectation. The remaining H_1 features concentrate in boundary regions with lower triangle density, consistent with structural gaps rather than random noise.

4 Results

4.1 Ranking Performance

We report two analyses with different time horizons. For **ranking evaluation** (RQ1), we use $H = 6$ years to reflect multi-year consolidation in HCI and report NDCG@50 with integration rate (the proportion of top-50 pairs that integrate within H). For **cycle-death analysis** (RQ2), we examine links that disappear by 2021 and use a 3-year post-disappearance window to classify outcomes as integration or decay, ensuring observations remain within our corpus through 2025.

Figure 2 summarizes the NDCG@50 results. H_1 links are competitive with structure-heavy baselines and stronger than similarity-only baselines in this low-evidence setting. The central point is not that H_1 maximizes accuracy, but that it remains competitive while returning traceable explanations: each link candidate is accompanied by a representative cycle (intermediate concepts) and evidence snippets, enabling inspection for why two areas are close but not yet merged. By contrast, similarity-based methods surface near neighbors without structural rationale, and triangle/link-prediction methods highlight near-closure edges that already sit inside dense neighborhoods. These signals are complementary rather than substitutes, suggesting that H_1 can be used to surface interpretable gaps

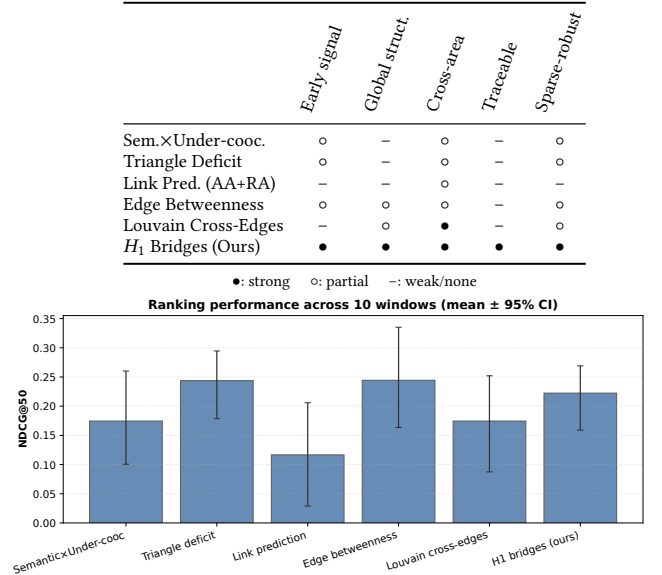


Figure 2: Cross-area link discovery methods: analytical capabilities (left) and NDCG@50 ranking performance across 10 windows (right). H_1 Bridges uniquely provides traceable explanations while remaining competitive with structure-heavy baselines.

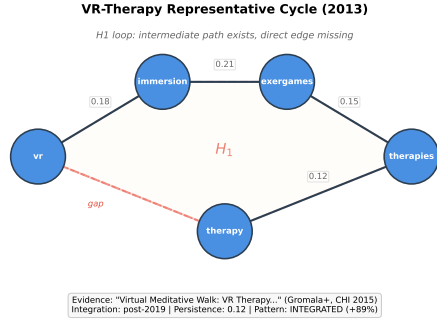


Figure 3: Representative cycle for VR–Therapy (2013). The H_1 loop shows semantic proximity through intermediates while the direct link is missing.

while standard baselines flag likely near-term closures. TDA provides a structural characterization rather than extrapolating volume-based trends by focusing on the emergence and cycle “death”: when triangles (2-simplices) form around a loop, the corresponding H_1 cycle becomes a boundary and disappears, which we interpret as the filling of a semantic gap through triadic closure. Lead time (first-hit) is measured for all methods via survival analysis with right-censoring at $T_{\text{end}} = 2025$. For H_1 links (2010–2019), the median lead time is approximately two years, and most links integrate within three years.

4.2 Case Studies and Explainability

We examined outcomes for links that disappeared by 2021. A disappearing link is classified as *integration* if co-occurrence exceeds the threshold within three years, and as *decay* otherwise. Many disappearing links followed the integration pattern, where links vanish as concepts become frequently co-occurring and triangles fill the structural gap. These results suggest that H_1 links serve as structural precursors to disciplinary integration. For each predicted bridge, we provide an *evidence snippet*: titles or abstract sentences from top-ranked papers (ranked by tf-idf relevance to the concept pair or intermediate chain) containing the endpoints and/or intermediate concepts. This allows researchers to validate whether a suggested bridge reflects a plausible conceptual mechanism.

We illustrate with a case study. “Therapy” was identified as a persistent boundary word spanning 1999–2019. In 2013, our method flagged “VR”–“Therapy” as a cross-area link candidate. The representative cycle computed from the Word2Vec embedding space was: $vr \rightarrow immersion \rightarrow exergames \rightarrow therapies \rightarrow therapy$, where each edge represents semantic proximity in the embedding space. This chain reveals a plausible bridging mechanism through exergaming applications. Representative papers such as “Virtual Meditative Walk: Virtual Reality Therapy for Chronic Pain” [14] exemplify the eventual integration. After the boundary period ended in 2019, concept frequency increased substantially (89%), consistent with the integration pattern in our classification. A contrasting case is Blockchain-HCI: a link formed between “Blockchain” and “Participatory Design” in 2018 but dissolved by 2021 as topics drifted apart, showing that not all detected links lead to integration.

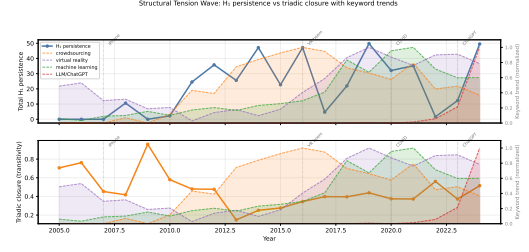


Figure 4: Structural tension wave with technology trend overlays (2005–2024). Blue lines show topological metrics: total H_1 persistence (top) and triadic closure (bottom). Shaded regions show normalized keyword frequency trends for four technology waves: crowdsourcing (orange, 2011–2016 peak), virtual reality (purple, 2016–2019 peak), machine learning (green, 2015–present), and LLM/ChatGPT (red, 2023–present). Vertical markers indicate external events (iPhone 2007, VR boom 2016, COVID 2020, ChatGPT 2023). Triadic closure (bottom; the fraction of connected concept triples that form complete triangles, indicating integration of indirect connections)

5 Discussion

Our results support the view that concept neighborhoods repeatedly become adjacent through intermediates before they become directly integrated in the literature. We interpret persistent 1-cycles as signals of adjacency without direct coupling, and we use loop stability as a proxy for how long these gaps remain unresolved. A loop indicates that a short mediated path exists between two endpoint concepts through intermediate concepts, while the direct co-occurrence edge (u, v) remains absent or negligible. This differs from frequency-based signals that summarize existing connections and from shared-neighbor predictors that prioritize near-term closure when many neighbors are already shared. By returning a representative cycle, loops provide a traceable explanation in the form of a concrete chain of intermediate concepts, which makes the suggested relationship inspectable rather than purely score-based.

Fig. 4 provides a macro-level summary of structural tension. Values remain near zero from 1999 to 2007, likely due to limited coverage and fewer papers in early years. A rise appears around 2008, consistent with rapid growth of mobile-related work creating new adjacencies not immediately integrated. From 2011 to 2016, tension stays high, consistent with sustained growth in wearables, sensing, and VR-related topics where areas approach each other through intermediates without immediate closure. Sharp drops, such as around 2017 and 2022, are consistent with our interpretation that gap closure corresponds to integration through triadic closure. Peaks around 2019 and 2024 are consistent with major external events (e.g., COVID, generative AI) bringing multiple areas into proximity before consolidation. Loops and standard link prediction methods operate at different decision horizons. Shared-neighbor predictors identify connections already strongly implied by existing neighborhoods and likely to close soon. Loops highlight missing shortcuts where endpoints are indirectly reachable through a short chain, yet direct coupling is absent. A practical workflow is to use

standard predictors for near-term monitoring and loops for horizon scanning of persistent cross-area gaps.

This perspective suggests concrete uses for researchers, committees, and community analysis. For researchers, the intermediate chain can act as a hypothesis generator that suggests what mechanisms, keywords, or prior work could plausibly bridge two areas. For program committees and funders, loop stability can help distinguish long-standing white spaces from transient co-mentions by prioritizing gaps that persist across years. For understanding how HCI evolves, accumulated structural gaps offer a compact diagnostic of when divergence dominates through gap accumulation versus when convergence dominates through gap closure. Limitations include static embeddings that can miss semantic drift, venue coverage that restricts claims beyond the selected conferences, and parameter choices that can affect which loops are detected. Polysemous terms like “interaction” may produce topological singularities that affect cycle detection [4]. Year-specific embeddings may undergo arbitrary rotations across years; we mitigate this by analyzing within-year structures and using persistence (threshold-invariant) rather than absolute distances, though direct cross-year comparisons remain limited. Identifying a loop signals a structural opportunity, not a guaranteed prediction; some gaps persist indefinitely due to fundamental differences between fields. Future work should incorporate temporal embeddings with alignment [21], expand the corpus beyond HCI venues, and evaluate usefulness with expert judgments.

AI Tools Disclosure

While LLMs assisted in research planning and coding, the authors generated all content and assume full responsibility.

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